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Integrated Workflow to Evaluate Completion and Stimulation Efficiency in a Carbonate Reservoir

Y. Huang and H. Zhao, ExxonMobil Technology & Engineering Company, Spring, Texas, United States; J. Mendez-Santiago, ExxonMobil Upstream Integrated Solutions Company, Spring, Texas, United States; R. Vijay and V. Sai, ExxonMobil Services & Technology Private Limited, Whitefield, Bangalore, India; M. Rashid Nasser Al Lamki, Exxon, Al-Khalij, Inc., Abu Dhabi, United Arab Emirates; F. A. Tavaréz, ExxonMobil Upstream Integrated Solutions Company, Spring, Texas, United States; D. Ibrahim, M. Ali Altamimi, A. Hassan Al Marzooqi, F. Khamis Alshamsi, and F. Jamal, ADNOC Offshore, Abu Dhabi, United Arab Emirates

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Abstract

Extended-reach horizontal wells have been drilled to develop a giant offshore carbonate field, with more infills planned, to increase daily production and enhance oil recovery. Well completion and stimulation strategies vary from one well to another and evolve as new technologies are implemented. How are these wells performing relative to expectations? An integrated workflow is developed to screen for Actual Completion Efficiency outliers and to guide stimulation prioritization and focus areas for completion design improvement.

Actual Completion Efficiency (ACE) is defined as the Productivity Index ratio

$$ACE = \frac{PI_{measured}}{PI_{baseline}} = \frac{\left(\frac{Q}{\Delta P_{reservoir} + \Delta P_{well}} \right)_{measured}}{\left(\frac{Q}{\Delta P_{reservoir} + \Delta P_{well}} \right)_{baseline}}$$

$\Delta P_{reservoir}$ is formation drawdown without skin, and estimated from a pressure buildup test, reservoir surveillance, or modeling. The same production rate Q for baseline and measured is usually chosen to avoid rate-dependency.

ΔP_{well} consists of wellbore friction, pressure drop through the completion, and skin. $(\Delta P_{well})_{baseline}$ represents ideal conditions without damage and is calculated using a well model. Ideally, $(\Delta P_{well})_{measured}$ is calculated from pressure measurements while the well is flowing.

In theory, any flow inefficiency within the wellbore and at the rockface would result in $(\Delta P_{well})_{measured} > (\Delta P_{well})_{baseline}$, therefore $ACE < 100\%$.

This workflow is demonstrated by evaluating ACE for unstimulated and stimulated wells consisting of long horizontal laterals completed with limited entry liner (LEL) in openhole.

For an unstimulated well, a pre-stim $PI_{baseline}$ assumes pristine completion and no formation damage. A well with low ACE number will be further investigated for impairment mechanisms (e.g., formation damage) and possibly prioritized for acid stimulation.

For a stimulated well, a carbonate acidizing model is developed to predict the target stimulated skin. Model capabilities include acid jetting through LEL and are calibrated with lab testing and field experiences. A post-stim $PI_{baseline}$ incorporated this target stimulated skin as an input.

Beyond a simple production uplift pre- vs. post-stimulation, the ACE workflow first establishes an expected post-stim $PI_{baseline}$, and then assesses whether stimulation target is met by comparing $PI_{measured}$ against it. The success of the ACE metric depends on the quality of the pressure and rate measurements, accuracy of the fluid properties, understanding of the subsurface along the lateral (e.g., permeability, porosity, lithology), and proper coordination with well operations for key data collection. Considering the practical input uncertainties and model simplifications, ACE values are compared on a relative basis among wells and interpreted in trend.

This workflow operationalizes the Actual Completion Efficiency definition into a practical metric to compare flow performance among wells. It provides a universal methodology to compare different completions, evaluate effectiveness of new technologies, and optimize stimulation operations. As a result, potential outliers can be identified, and new insights can be obtained regarding key factors that drive improvements in completion and stimulation design.

Introduction

A giant offshore carbonate oil field has been under development for over 30 years with evolving new technologies in drilling and completion. Extended reach wells have been drilled from artificially constructed islands to maximize reservoir contact. Many of these wells are completed with limited entry liner (LEL) in long open-hole laterals, particularly in tight reservoir. LEL completion has facilitated acid placement along the whole wellbore, enabling bull heading operations, instead of more complex and costly coil tubing stimulation. During recent years, a few ultra-extended reach wells have been drilled with over 30,000 ft laterals [Panga *et al.* 2024]. New technologies and best practices have been implemented to optimize operations and deliver wells with high productivity.

While striving to enhance operational efficiency, it is also crucial to evaluate the wells' performance. This includes assessing the impact of drilling and completion on well performance to identify and guide focus areas for improvement. The initial step is to define a unified metric that enables relative comparison of completion performance, while accounting for variations in fluid PVT properties, reservoir quality, and completion type. This ensures a consistent evaluation across wells, facilitating meaningful comparisons of completion strategies.

Production (or injection) rate may be the most straightforward and commonly discussed metric to assess well performance. However, rate is not only an indication of well performance, but also impacted by production target, surface equipment constraints and/or well operability limits, etc. To minimize the impact of the above factors, Productivity Index (PI, the ratio of rate over drawdown) can be utilized to assess the deliverability of a well. To compare between wells, one step further is to normalize PI, by reservoir and fluid properties like viscosity, formation volume factor, pay zone thickness, completion length [Levitan *et al.* 2004, Konwar *et al.* 2023]. PI is usually normalized linearly by these parameters, assuming most of the drawdown is within the formation, given the wellbore pressure drop is generally non-linear to rate.

Skin, representing impairment pressure drop (or reduced dP for stimulation) normalized to the ideal drawdown, could be a completion focused metric. However, skin value is often a source of confusion in practice, due to its many inconsistent definitions. While a well engineer may associate drilling and completion damage with skin, well testing engineers may interpret Pressure Transient Analysis (PTA) skin to also include wellbore friction, well angle and partial penetration. Some reservoir simulators may imply

the skin input value based on radial Darcy equation regardless of the well angle, while a nodal analysis tool may (should) have dedicated horizontal PI analytical models that requires different normalization for skin.

The well length in the analytical PTA for horizontal well may be perceived as an effective contributing length for flow. However, this combines the impact from the reservoir heterogeneities along the well and in the near well region, and rock face impairment.

In summary, although the above metrics provide various perspectives and insights about well performance, they are not intended for assessing completion-related flow efficiency and to enable comparison among wells. Therefore, a practical workflow based on the Actual Completion Efficiency is developed in this work.

Method

Pre-stim ACE Workflow

Starting from its textbook definition, the Actual Completion Efficiency (ACE) is calculated as the ratio of $PI_{measured}$ over $PI_{baseline}$.

$$ACE = \frac{PI_{measured}}{PI_{baseline}} = \frac{\left(\frac{Q}{\Delta P_{reservoir} + \Delta P_{well}} \right)_{measured}}{\left(\frac{Q}{\Delta P_{reservoir} + \Delta P_{well}} \right)_{baseline}} \quad \text{Eq. 1}$$

The scope of current workflow is applicable to conventional non-fractured completions. To assess the initial completion efficiency, the optimal time window should be an early flow period after well startup while production is also stabilized. During the early flow period, the rockface and wellbore is susceptible to minimal additional damage except from drilling and completion operations, therefore the $PI_{measured}$ is most accurate for completion efficiency assessment. The same production rate Q for baseline and $PI_{measured}$ is usually chosen to avoid rate-dependency effect.

$PI_{measured}$ has a straightforward definition and is calculated from production surveillance data.

$$PI_{measured} = \frac{Q}{P_{reservoir} - FBHP} \quad \text{Eq. 2}$$

The flowing bottomhole pressure $FBHP$ is obtained from the downhole gauge, if available. The reservoir pressure $P_{reservoir}$ may be inferred from a pressure buildup test (or a falloff test for injector), inflow performance curve [Ismadi *et al.* 2010], stethoscope measurement, or pre-startup bottomhole pressure.

$PI_{baseline}$ assumes the well has no impairment and is calculated using a well productivity model (commercial software is used). It includes the following:

- Formation dP, i.e. from well drainage to rockface. It may be estimated from PTA, if well test is available. Or estimate with a PI model using reservoir permeability, fluid PVT, drainage distances, and well geometry, etc.,
- dP from completion limited entry liner if any, based on pressure gauge readings, if available, or from correlations, and
- Tubing friction and hydrostatic head from the toe to the downhole gauge.

$PI_{baseline}$ assumes the entire completion length contributes to flow and no near-well formation damage, i.e. 0 skin for un-stimulated well model. In other words, $PI_{baseline}$ does not include:

- Formation damage from drill-in fluid invasion, plugging from migrated sand after initial production, or asphaltene deposition in the near well formation,
- Liner nozzle plugging due to scale, solid, or asphaltene, or

- Additional tubing/liner friction, or flow area reduction due to asphaltene or scale buildup.

For a carbonate well that is not acid-stimulated during initial completion, the *pre-stim ACE* may be assessed as outlined above, where $PI_{baseline, pre-stim}$ assumes skin 0, i.e. no impairment and no stimulation (Well Productivity Model 1 in Fig. 1).

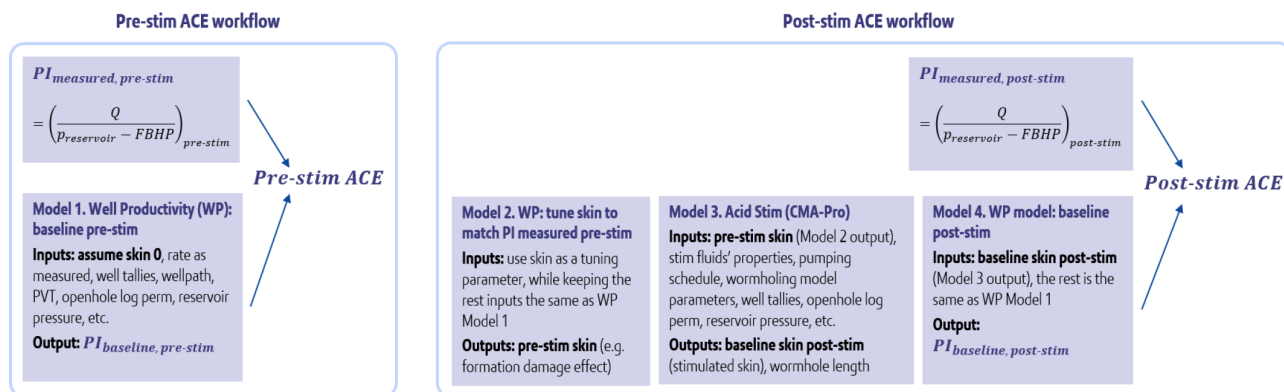


Figure 1—Pre-stim and post-stim ACE workflow flowchart. Well productivity models utilize commercial software. Acid stimulation model (referred as CMA-Pro) is developed in house.

Post-stim ACE Workflow

To evaluate matrix stimulation efficiency in carbonate formation, a post-stimulation Actual Completion Efficiency (*post-stim ACE*) workflow is developed (Fig. 1). *Post-stim ACE* is still calculated as the ratio of $PI_{measured}$ over $PI_{baseline}$, but with production data *after* stimulation, and with acid treatment inputs, respectively. Notice that $PI_{baseline, post-stim}$ includes stimulation skin's effect (negative skin from acid stim model), therefore an enhancement from $PI_{baseline, pre-stim}$ (skin 0).

Proprietary software is developed for LEL completion design to optimize acid stimulation effectiveness [Sau *et al.* 2014a; Sau *et al.* 2014b; Hosani, *et al.* 2016]. The software, Carbonate Matrix Acidizing and Productivity (abbreviated as CMA-Pro from here onwards) models transient flow from the surface to the reservoir. The resulting fluids placement and stimulation profile (wormhole growth and stimulated skin) are predicted along the completion. The software provides an integrated methodology to model LEL completion including the acid jetting effect from the small nozzles in the liner along open-hole laterals.

CMA-Pro requires a pre-stim skin as input. It may be estimated from tuning a well productivity model to match $PI_{measured, pre-stim}$ (Well Productivity Model 2 in Fig. 1). Then the CMA-Pro model is run (Acid Stim Model 3 in Fig. 1) and is calibrated to actual treatment data. CMA-Pro model outputs (predicts) the baseline post-stim skin. Finally, this baseline post-stim skin is input into Well Productivity Model 4 in Fig. 1 which then calculates $PI_{baseline, post-stim}$. This $PI_{baseline, post-stim}$ effectively sets a target for successful acid treatment for this well.

Post-stim ACE then compares $PI_{measured, post-stim}$ the measured well productivity enhanced by wormholes against $PI_{baseline, post-stim}$ predicted by Well Productivity Model 4 in Fig. 1.

Data Quality for Well Models Input

For consistent ACE assessment, quality inputs are paramount to sound well productivity and acid stimulation modeling. A few suggestions are below.

- Well completion: Joint lengths and inner diameters are critical for friction calculation. Verify proper use in the well models.
- Production data: To assess initial completion efficiency, target an early and stable flow period after the well is cleaned up. If a shut-in well test is arranged, coordinate with operations on data

acquisition best practice: hard shut-in, gauge pressure recorded with good temporal resolution, and minimizing nearby well activities.

- Subsurface: Consult the reservoir engineer on proper nearwell permeability for well productivity modeling (petrophysical permeability log, well test kh, reservoir model permeability, etc.). For well drainage area and effective pay thickness, infer distance to boundaries (well test, seismic map), well interference, etc.

Results and Discussion

The pre-stim and post-stim ACE workflows have been utilized to assess the completion efficiency of several wells with LEL completion. The results have been tied to drilling/completion operations and stimulation treatments, providing valuable insights into how successful the drilling, completion and stimulation operations are. A well with relative low *pre-stim ACE* may warrant further investigation, such as excess damage from mud or completion brine loss, ineffective coil tubing cleaning, etc. If formation damage is the concern, this well may have higher potential for production enhancement from acid stimulation. The *post-stim ACE* is a metric to assess acid treatment outcome. A relative low *post-stim ACE* may indicate that the acid treatment can be optimized to improve stimulation efficiency.

Case Study – Well A

Well A is an extended reach horizontal oil producer with primarily 6-5/8" LEL completion. This LEL section is compartmentalized with swellable packers. Due to operational challenges, a 5-1/2" tapered pre-perforated liner without a packer was placed at the end of the 6-5/8" liner, with the toe section left as barefoot openhole (Fig. 2). Well A was not stimulated during initial completion.

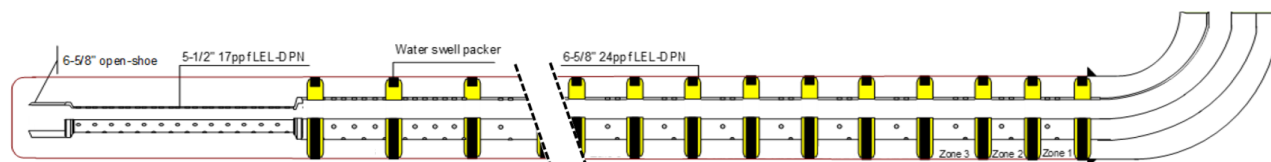


Figure 2—Well A completion scheme

Well A Pre-Stimulation ACE Evaluation

The pre-stimulation ACE workflow compares the measured well productivity to its baseline potential, helping to identify underperformers and prioritize stimulation candidates.

The evaluation for Well A followed a structured approach:

1. Baseline Well Performance Model

A pseudo-steady state well model was built incorporating key reservoir parameters such as permeability, fluid PVT data, and relative permeability curves, etc. The model assumed pristine completion with zero skin to estimate the baseline PI representing a well productivity target under undamaged condition.

2. Measured PI from Well Test

$PI_{measured}$ was derived from an early well test conducted under stable flow condition after initial completion.

3. ACE Calculation

ACE was calculated as the ratio of the $PI_{measured}$ over $PI_{baseline}$. For Well A, *ACE* was approximately 81%, indicating that the well may underperform relative to its potential. The relatively low *ACE* suggested that this well may be a good candidate for stimulation.

Well A Post-Stimulation ACE Evaluation

To further quantify the under-performance of initial completion, the well productivity model was calibrated by tuning the skin until the modeled PI matched $PI_{measured}$. A positive skin of +1.2 was required. Note that calculating pre-stim skin is not required in pre-stim ACE workflow, however it is an input to acid stimulation model for post-stim ACE workflow.

Following the identification of Well A as a candidate for stimulation, an HCl treatment was executed.

To assess the HCl treatment effectiveness, an acid stimulation model was then built to replicate the treatment and match surface pressure responses observed during the treatment. Key operation data, including injection rates, surface pressure, acid volumes, and fluid properties, were included in model inputs. The pre-stim skin +1.2 was assigned as the initial skin before acid stimulation. Sensitivity analysis was performed to understand the impact of key parameters on the treatment pressure. The model was iteratively refined until a reasonable match was achieved between the modeled and actual surface pressure recorded during the job (Fig. 3).

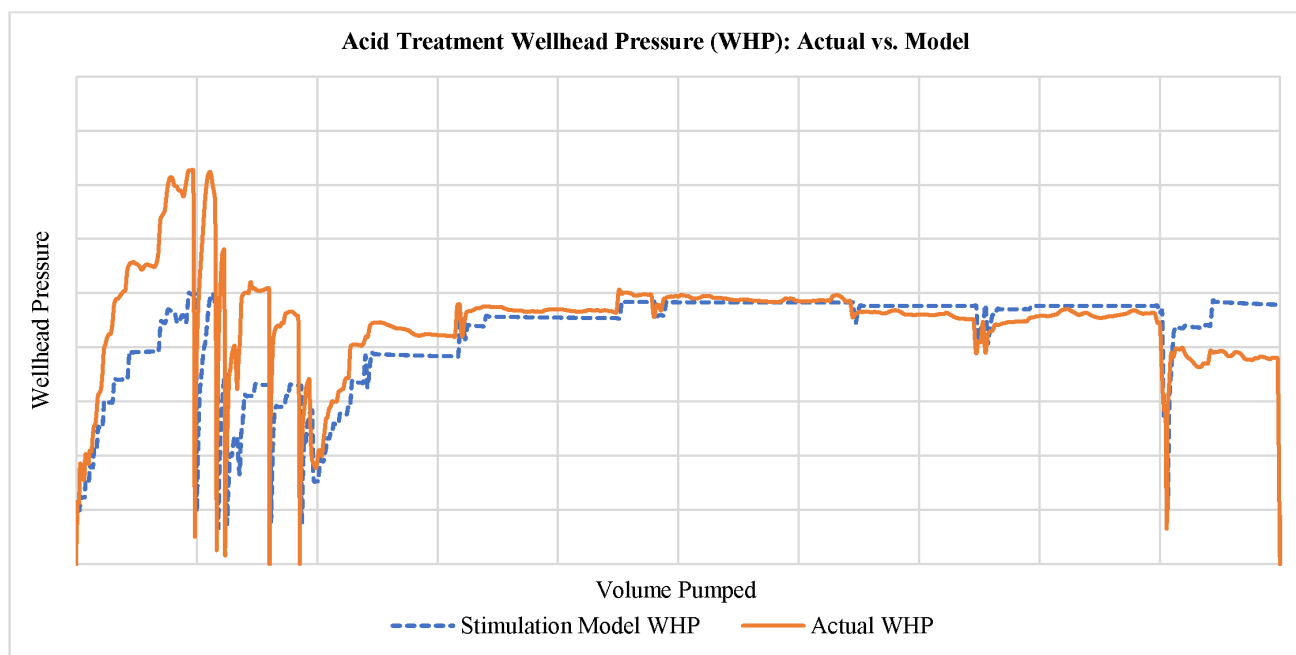


Figure 3—Pressure match results between modeling and field data during acid treatment for Well A

The acid stimulation model predicted -2.1 post-stim baseline skin which would be a significant reduction from $+1.2$ pre-stim skin, if effective stimulation is achieved (Fig. 4 Fig. 3). The acid stimulation model also predicted an average wormhole length of 1.6 feet which would be considered effective penetration beyond the rock face. This analysis provided insights into the potential stimulation outcome and supported the treatment design.

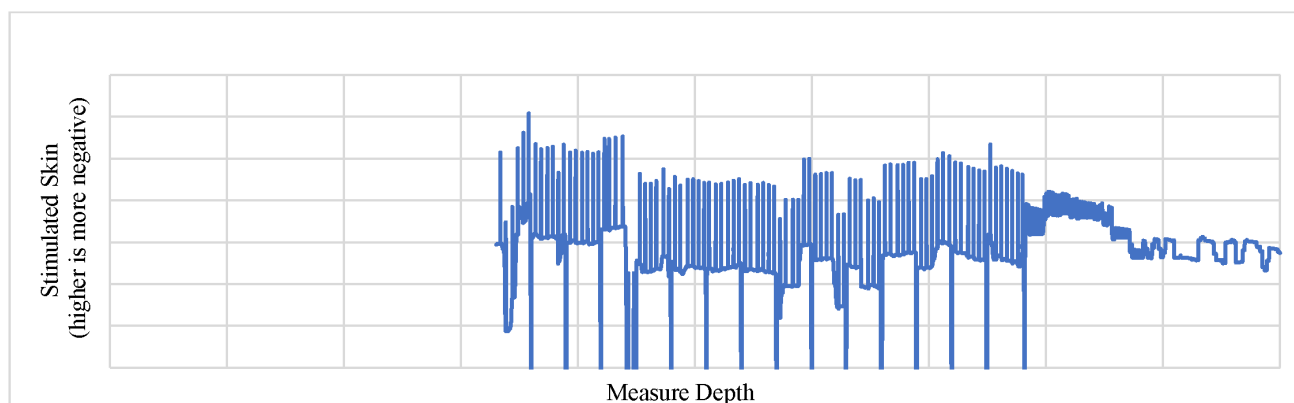


Figure 4—Stimulated skin from acid stimulation modeling for Well A

Following the acid stimulation modeling and pressure match exercise, the final step was to quantify the effectiveness of the stimulation treatment relative to the expectation. The well productivity model for Well A was updated with the post-stimulation skin -2.1 (predicted from the acid stimulation model) and calculated the $PI_{baseline, post-stim} \cdot PI_{measured, post-stim}$ is calculated using an actual well test and production data. $PI_{measured, post-stim}$ was 1.15 times of $PI_{baseline, post-stim}$, i.e. **post-stim ACE** at 115%. A **post-stim ACE** value over 100% may suggest that acid stimulation exceeded expectation.

This case study is an example of the integrated modeling workflow in quantifying treatment outcomes. The ability to reconcile modeled and actual performance provides a framework for optimizing future stimulation designs in analogous carbonate reservoir environments.

Case Study – Well B

Well B is also a horizontal oil producer well with 6-5/8" LEL in the lower completion (Fig. 5). The whole LEL section is divided into compartments with packers. Due to relatively high reservoir permeability, wells in this area are usually stimulated later in life when the production rates decline. Well B has not been stimulated as of writing. Nevertheless, **pre-stim ACE** is evaluated for initial completion efficiency.

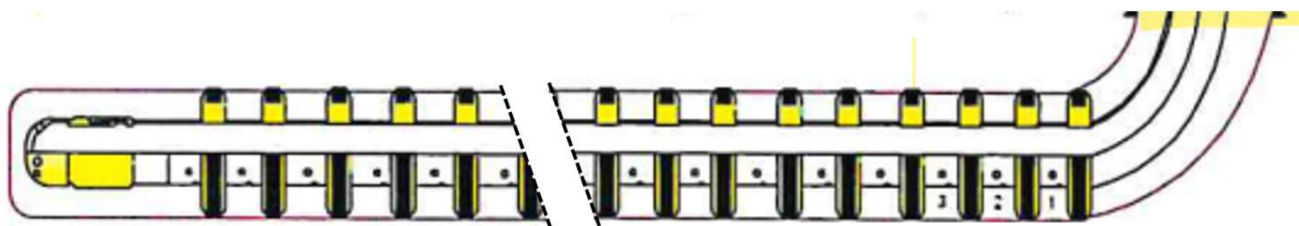


Figure 5—Well B completion scheme

Well B Pre-Stimulation ACE Evaluation

Well B's initial completion efficiency was assessed for an early stable flow period, using the pre-stim ACE workflow. Its **pre-stim ACE** was determined to be approximately 103%, indicating that the well was producing at or slightly above its baseline potential. This finding suggested minimal initial completion damage, therefore Well B was not prioritized for early stimulation.

Note that $PI_{baseline}$ predicted from the baseline well productivity model is subject to input uncertainties (e.g. reservoir permeability, production rate metering, etc.) and well productivity model simplifications. Meanwhile, despite $PI_{measured}$'s straightforward definition, it is also affected by production rate accuracy and local reservoir pressure estimate. All above input uncertainties would introduce some variation in ACE value (In Well B example, calculated **ACE** is over 100%). Rather than over-reading into individual well's

absolute ACE value, ACE values should be compared among similar wells and interpreted on a relative basis. ACE metric is a screening tool to flagging outliers for further investigation, identifying stimulation candidates and optimizing resource allocation across the field.

Summary

The ACE workflow provides a single metric for relative comparison of completion performance, while accounting for differences in fluid PVT, reservoir quality, completion type, etc. The goal is to have a better understanding of how design and execution practices impact well performance, leading to early identification of improvement opportunities that can be applied to future wells in either the design and/or execution stage. The workflow may also help identify wellwork opportunities to improve productivity and improve future completion design to maximize the value of the asset via higher recovery per well.

As a leading indicator, ACE screens for well performance outliers, and would require further diagnosis to identify impairment mechanism. When comparing ACE values, interpret with consideration to input uncertainties and model simplifications. Focus on trends, rather than absolute values, given the expected uncertainty in the analysis.

Despite ACE's simple definition, the success of the ACE metric depends on:

- Alignment across related disciplines (wells, reservoir, and production, etc.): Strong collaboration with the objective to assess well performance for improvement. Exchange on methodologies and learnings.
- Quality data collection and proper interpretation: Coordinate with operations on data collection program. Ensure proper reservoir dP and wellbore dP estimates in the context of ACE calculation.
- Standardize workflow and support team in place: Develop a wellbore focused model to break down completion pressure drop. Take a practical approach for simple concept and consistent assessment over many wells.

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